

TEACHING MASTERCLASS — micro2dfn

From pixels to predictions: reading an anode cross-section honestly

The object: a thin graphite+silicon coating — the negative electrode of a lithium-ion battery — sliced open and photographed.

How that photo becomes material labels, how labels become quality-control comparisons and model inputs, and exactly where the evidence stops — worked end-to-end on real data.

Data: 34 electron-microscope fields of view · image-level comparisons only · all numbers traceable via [claim_evidence.csv](#)

Two routes through the material

— Fast route — 10 slides, ~15 min (marked ROUTE)

- title → the question → one image → the two systems → the frozen ruler
→ measurements → the one supported finding → the model's answer →
where evidence stops → take-home

— Teaching path — full deck + appendix

- worked image transformations, statistics, reconciliation, model mechanics,
failure modes, roadmap

— Companion files

- presenter_script.md — verbatim narration
- metric_dictionary.csv — every metric: units, meaning, limits
- claim_evidence.csv — every number → its source file
- roadmap.md — what a prospective validation would require

Three batches, three new photos, one question

— The setup

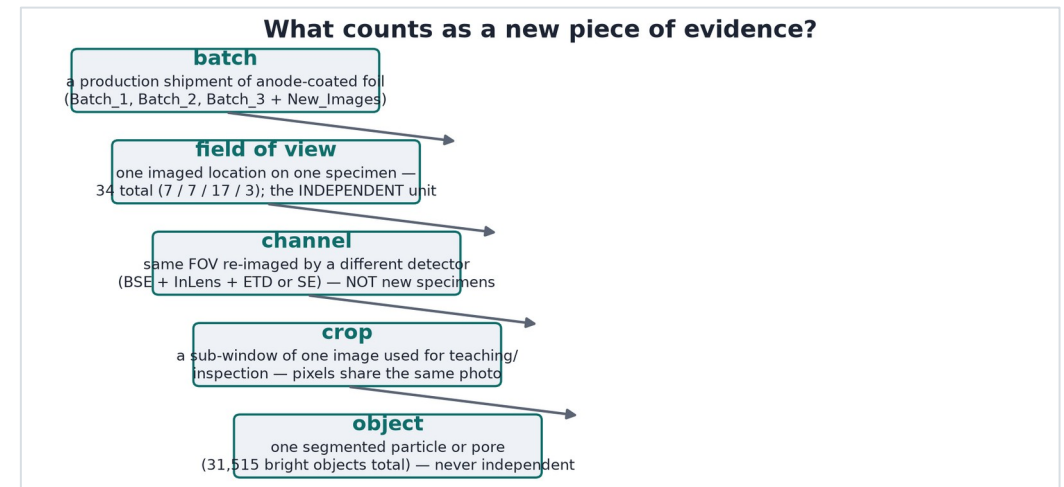
- Thin graphite+silicon electrode coatings, sliced open (ion-milled — polished flat with an ion beam) and imaged by SEM back-scattered electrons (BSE).
- Batch_3 = the shipment-approved reference. Batch_1 and Batch_2 = candidates. Plus 3 unlabelled new photos.

— The practical questions

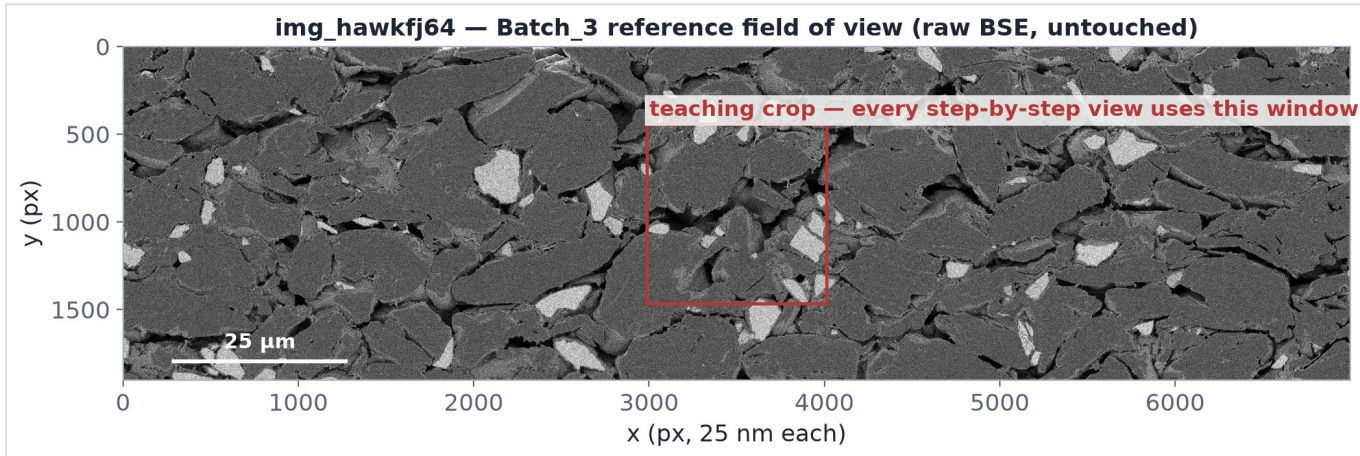
- Do the candidate batches look like the reference? (the quality-control, QC, job)
- Where they differ, what could it mean downstream? (the consequence job)

— What this deck will NOT claim

- no chemistry, no measured cell performance, no lifetime, no safety verdicts — images cannot supply those



One field of view: what BSE brightness means



— Reading the greys

- near-black = open pore / resin
- mid-grey = graphite + binder bulk
- bright = silicon-family (higher mean Z)

— The worked example

- img_hawkfj64 — the Batch_3 image nearest the batch median on all 9 declared features (a documented selection rule, not cherry-picked)

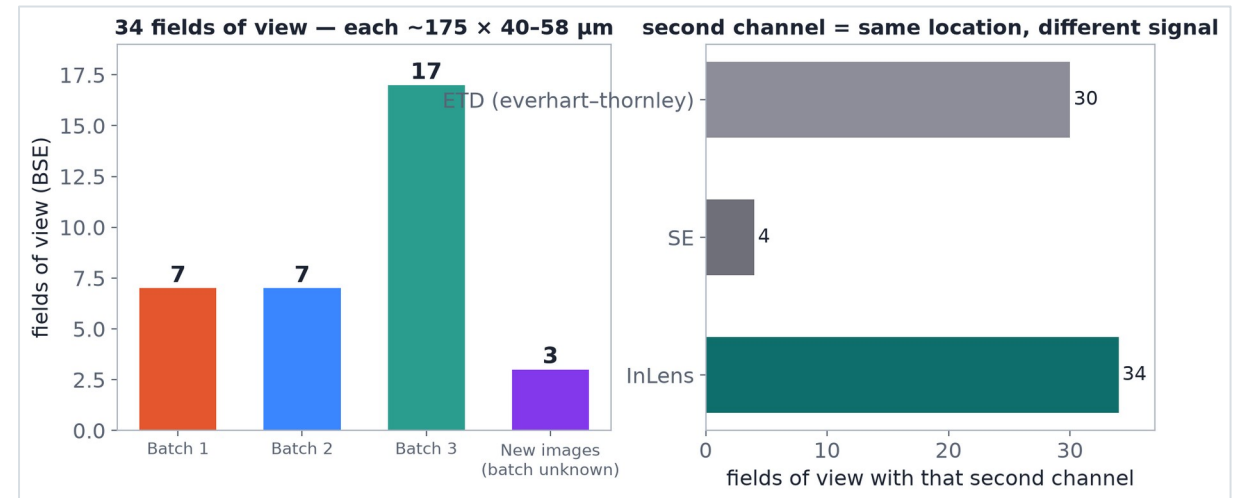
— 25 nm per pixel

- a sampling pitch, not a resolution — the effective object floor is $\sim 0.13 \mu\text{m}$ pores / $\sim 0.35 \mu\text{m}$ Si
- scale check: a human hair $\approx 70 \mu\text{m}$ — the smallest pores we can see are $\sim 1/500$ th of that

THE DATA

The dataset: 34 fields of view, three channels

- **Batch_1** — 7 fields of view (FOVs)
- **Batch_2** — 7 FOVs
- **Batch_3** — 17 FOVs (the reference)
- **New_Images_Batch** — 3 unlabelled FOVs
- **Channels**
 - BSE: all 34 — the segmentation channel
 - ETD (Everhart–Thornley, surface-sensitive): 30
 - InLens (topography): all 34
- **Images are the unit**
 - 34 is small; image-to-image variance sets every bound below



What these images cannot tell us

— Below the object floor

- sub-resolution porosity (~ 0.15 – 0.30 fraction exists but is invisible
— enters the model only as an ASSUMED add-on)

— Not in a 2-D section

- 3-D pore connectivity, true through-plane tortuosity
- electrode thickness (foil never in frame $\rightarrow$ assumed $75\text{ }\mu\text{m}$)

— Not in BSE at all

- binder distribution, the SEI film (the thin protective layer that forms on electrodes), lithium inventory
- chemical identity of bright material $\rightarrow$ needs EDS (elemental spectroscopy)

— Not in a photo

- measured capacity, cycle life, safety

— Consequence for honesty

- every model input that is not measured is declared and swept over a band (assumption_table.csv)
- the wall between measured and assumed is drawn on every slide from here on

Same pixels, two different jobs

— polaron_qc — anomaly detection

- question: does this batch break the reference's own envelope?
- frozen baseline thresholds → robust z (a 'how unusual' score) → fingerprints → verdict (ACCEPT / INVESTIGATE / REJECT)
- verdict = similarity to baseline, not material truth

— micro2dfn — measurement → consequence

- question: what structure is resolved, and what could it do in a physics model?
- shared segmentation → OBJECT classifier → ~30 markers → a battery-physics simulation (PyBaMM DFN)

— They share the ruler

- same flat-field, same multi-Otsu idea, same pixel scale

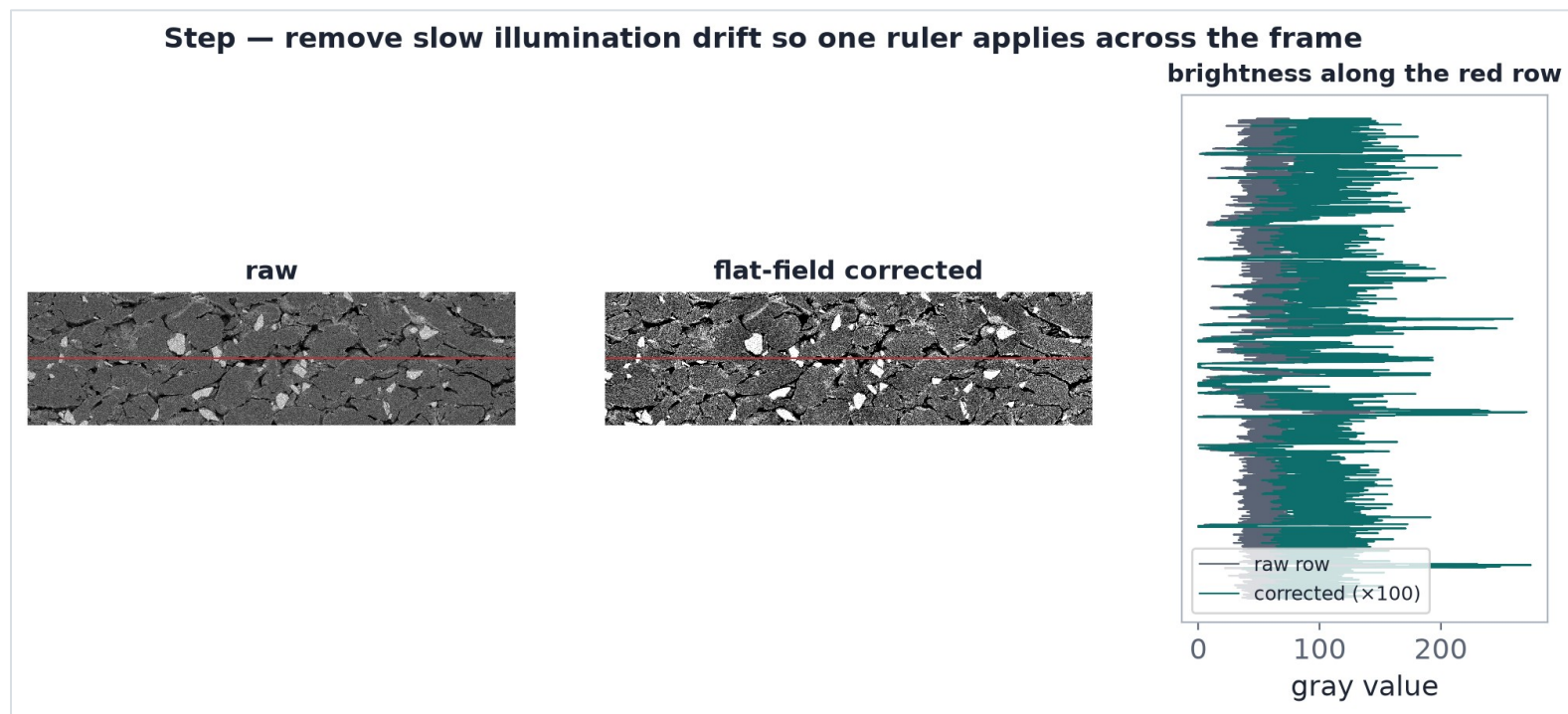
— They differ at ONE decision

- what counts as a silicon PARTICLE vs ambiguous bright material — and that single difference explains every downstream disagreement ($r = 0.27$ on classified Si, 0.8–1.0 on everything else)

— No claim of which is 'better'

- accuracy needs common ground truth — which does not exist yet; see roadmap.md

Step 1: remove the illumination gradient



— The problem

- large-area intensity drift (beam, detector, charging) makes one global threshold unfair

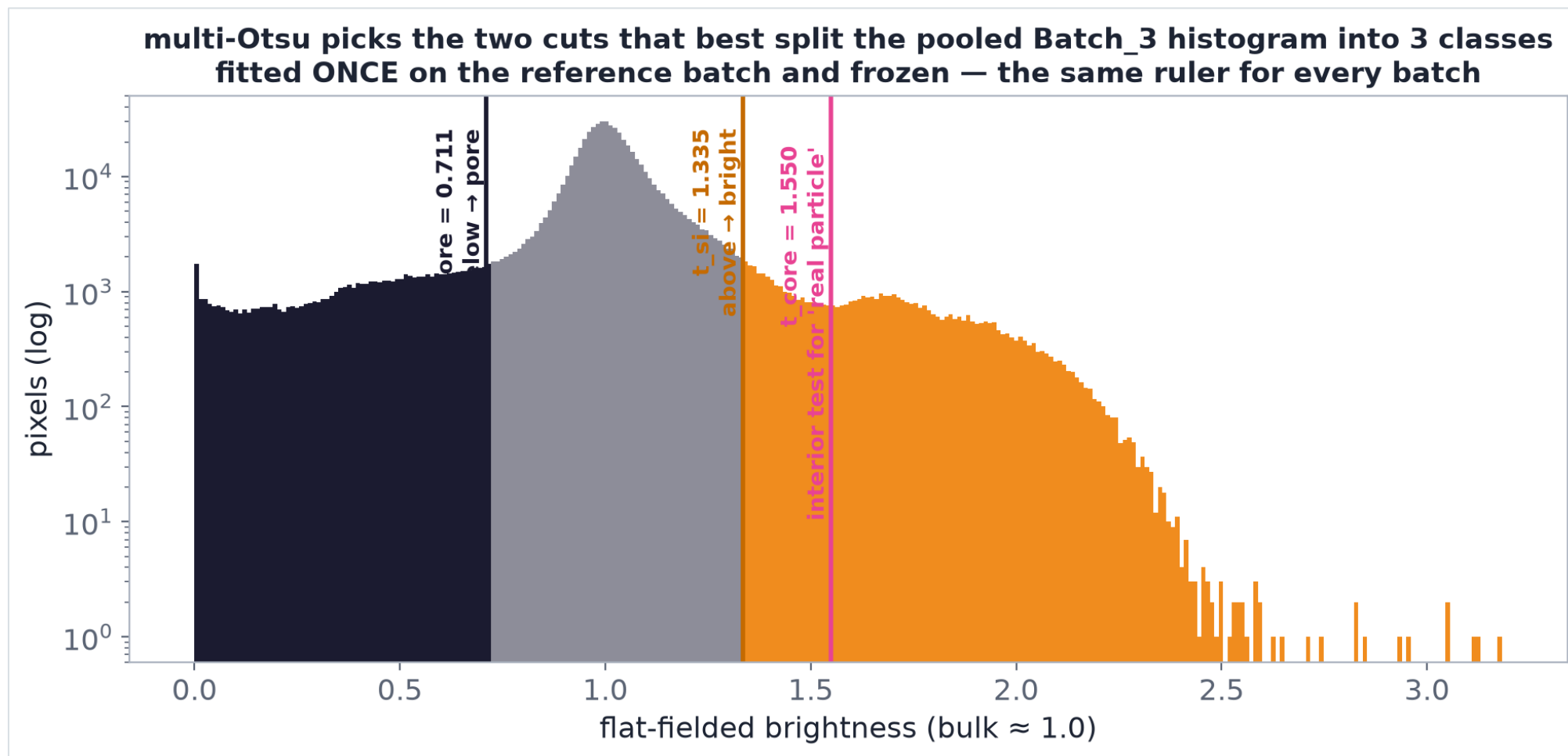
— The fix

- divide by a heavily-smoothed copy of itself (Gaussian sigma-200 on a 16x downsample)

— What stays honest

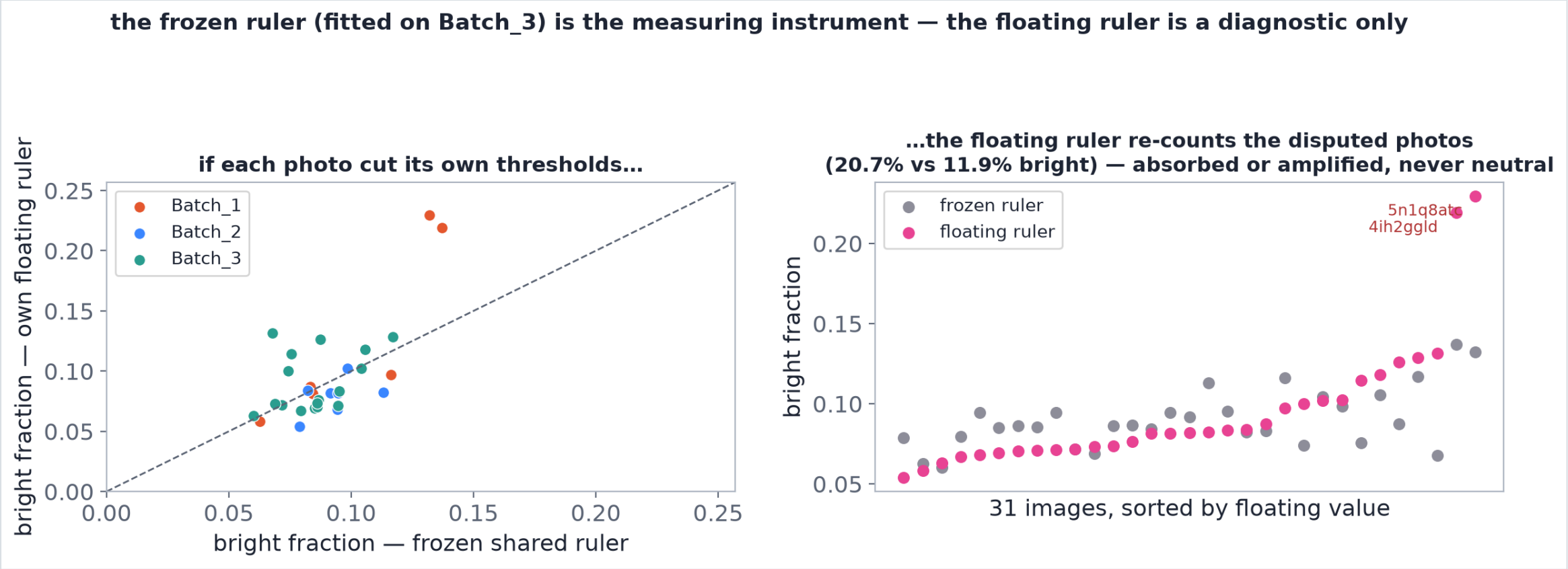
- thresholds are still fit on the ORIGINAL intensities of the reference — the flat field only de-trends

Step 2: the histogram is where labels come from



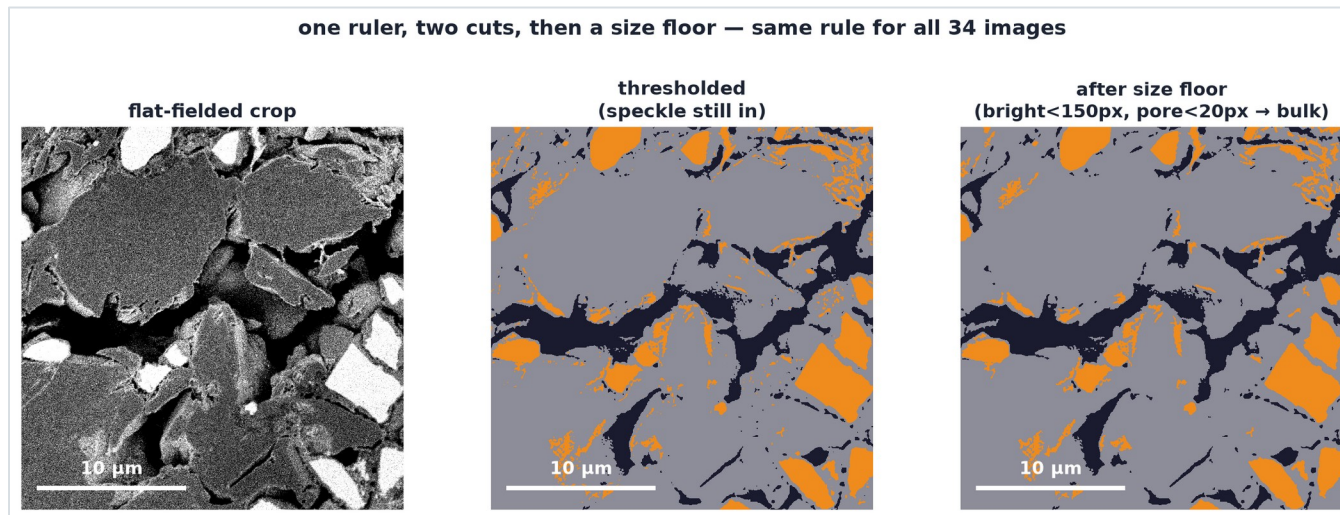
Multi-Otsu finds the two valleys that split pore / bulk / bright. Thresholds are RECIPE CHOICES — different choices, different labels, different findings.

Step 3: freeze the ruler, or it absorbs the anomaly



Frozen thresholds are fit ONCE on Batch_3 pixels and reused on every image. A per-image (floating) threshold re-centres on each image and can hide exactly the shift we are looking for.

Steps 4–5: threshold, then split touching objects



— Thresholding gives blobs

- but adjacent particles fuse into one object

— Watershed splits on the distance transform

- peaks inside the blob become seeds; boundaries flood between them

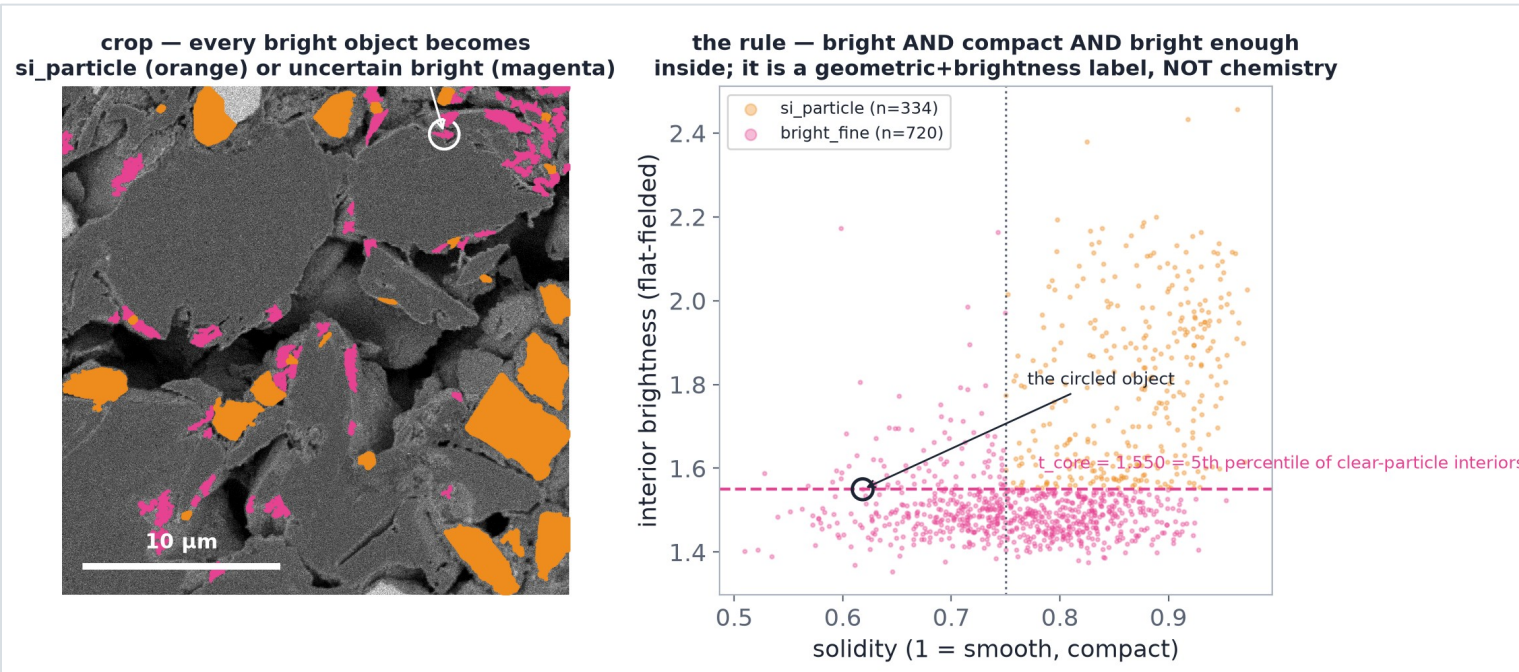
— Why we must split

- particle size and clustering metrics are meaningless while objects are welded

— The cost

- over-splitting is possible → size metrics carry a splitting bias (a stated limitation)

Step 6: the object classifier — where the two pipelines part



Three labels for bright objects

- si_particle — interior $\geq t_{core}$ AND solidity ≥ 0.75 (solidity = how filled-in the shape is: area $\div$ tight-fit outline area)
- bright_fine — fails the cut: small, low-solidity, or low-contrast ambiguous material
- rim/edge — bright halo at a pore edge (touched-border crescents)

Why it exists

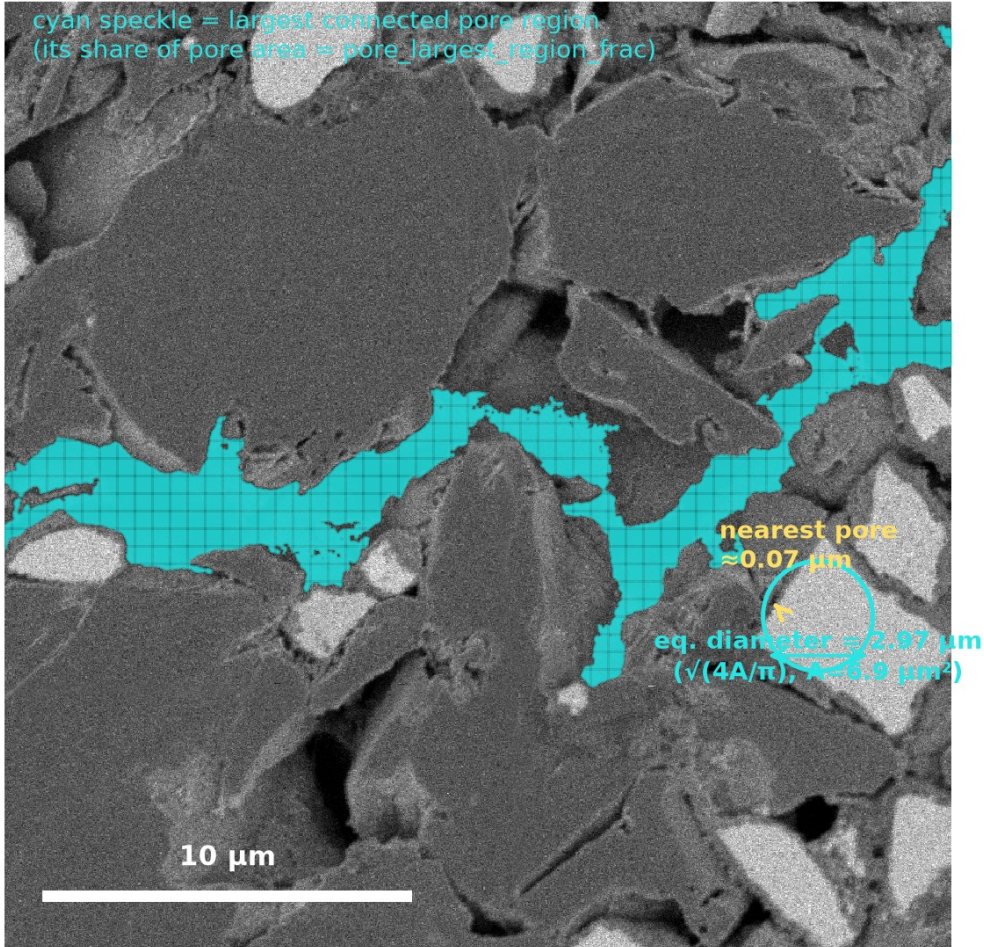
- the DFN needs PARTICLES; rims and dust are not particles

Why it matters later

- this is the ONLY metric where the two pipelines disagree ($r = 0.27$)

Step 7: objects become measurements

img_hawkfj64 crop — worked measurements (agent annotations over raw pixels)



— Per object

- eq. diameter = $\sqrt{4A/\pi}$; solidity; orientation

— Distances

- distance transform from pore mask → Si-to-pore proximity (a 2-D wetting proxy)

— Topology

- largest connected pore region share — network fragmentation

— Then aggregate

- per-image summaries (medians, area-weighted D50) — the image stays the unit

~30 markers, each with units, limits, and a job

— Every marker declares

- units · equation · worked number · what an increase means
- limitations · recipe-sensitivity · where it feeds the model

— Classes (the honesty axis)

- image-derived — measurable from pixels (pore fraction)
- geometry-derived — computed on objects (D50, Clark-Evans)
- physical proxy — heuristics (Bruggeman exponent)
- assumed input — declared, swept (thickness, sub-res porosity)
- simulation output — model results (stress, capacity)

— The watchlist

- 8 metrics that carry the whole interpretation — one-page watchlist in the repo

— Watchlist (grouped)

- acquisition guards: Si:bulk contrast, noise_MAD (image-noise meter), sharpness
- resolved pore fraction (reliable, DFN-adjacent)
- pore count + noise-adjusted count (the supported finding)
- Si candidate + uncertain-bright fractions (report TOGETHER)
- Si size: area-weighted D50/D90 → DFN radius
- Si → pore proximity + low-coverage share
- model stress + swelling budget (assumption-dependent)

Comparing batches: image is the unit, recipe is frozen

— The design (validated_comparison/recipe.json)

- fit thresholds + noise model on Batch_3 ONLY, freeze, apply to all 31 images
- summarise each image to 9 declared features
- compare batches on difference of medians, image unit

— The tests

- exact permutation — every label assignment enumerated: $C(24,7)=346,104$ (B1–B3), $C(14,7)=3,432$ (B2–B3)
- Benjamini–Hochberg FDR 0.10 — a correction that caps the expected share of false 'significant' hits at 10% across the 9 tests per pair
- bootstrap confidence intervals, seed 0; a leave-one-out (LOO) sweep for fragility

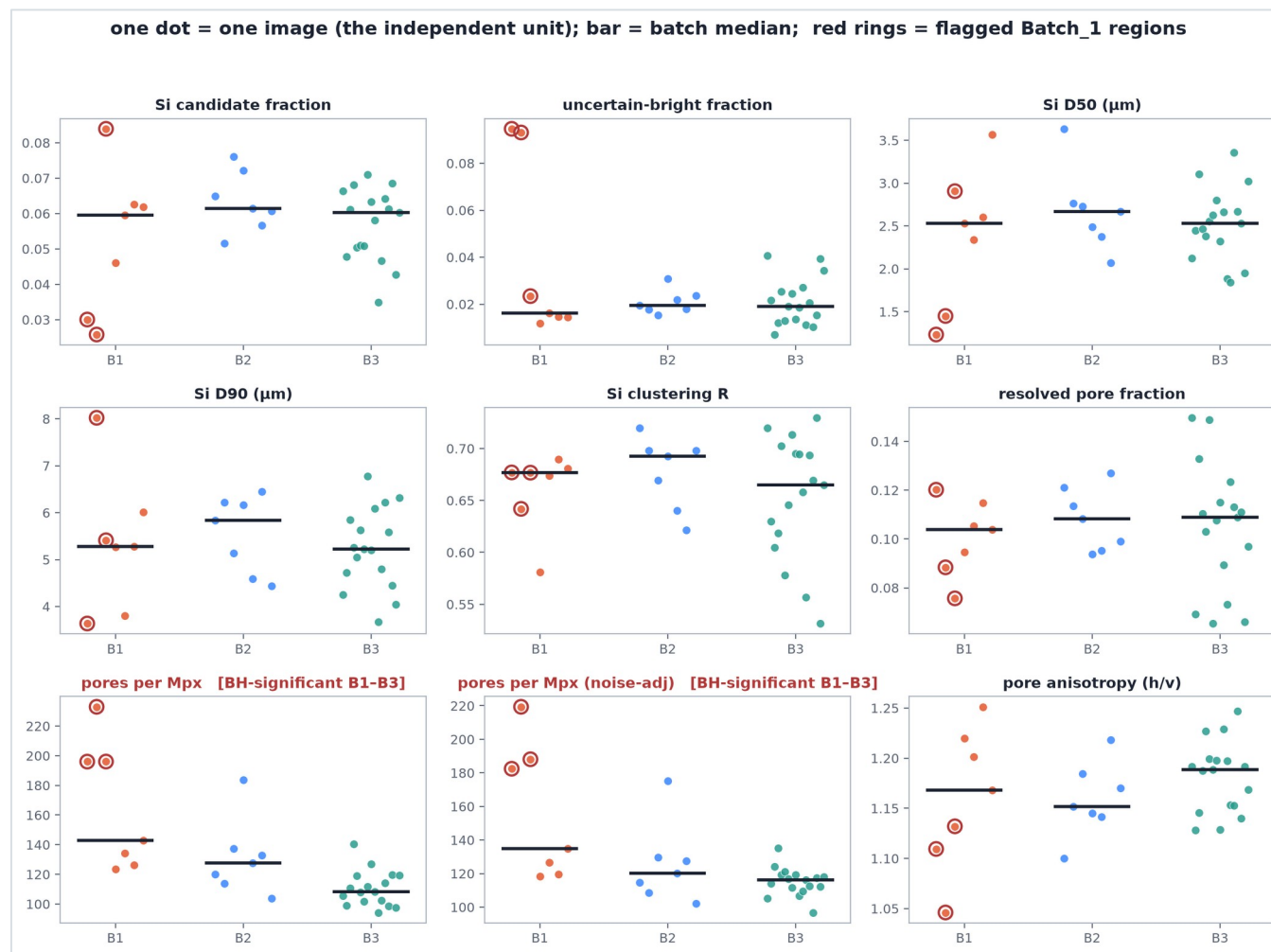
— And for every null result

- a minimum detectable difference (MDD) is reported — 'no clear difference' != equivalence

— Why so defensive?

- the earlier QC run claimed 'zero baseline false alarms' — the corrected LOO calibration measures $1/17 \approx 5.9\%$ [CI 1.0–26.9%]
- every safeguard below exists because a check FAILED once

The result: one supported difference out of nine



Supported (BH-significant)

- Batch_1 vs Batch_3: resolved pore count +34.5/Mpx, exact $p = 0.0005$
- ... and after noise adjustment: +18.8/Mpx, $p = 0.002$

Everything else

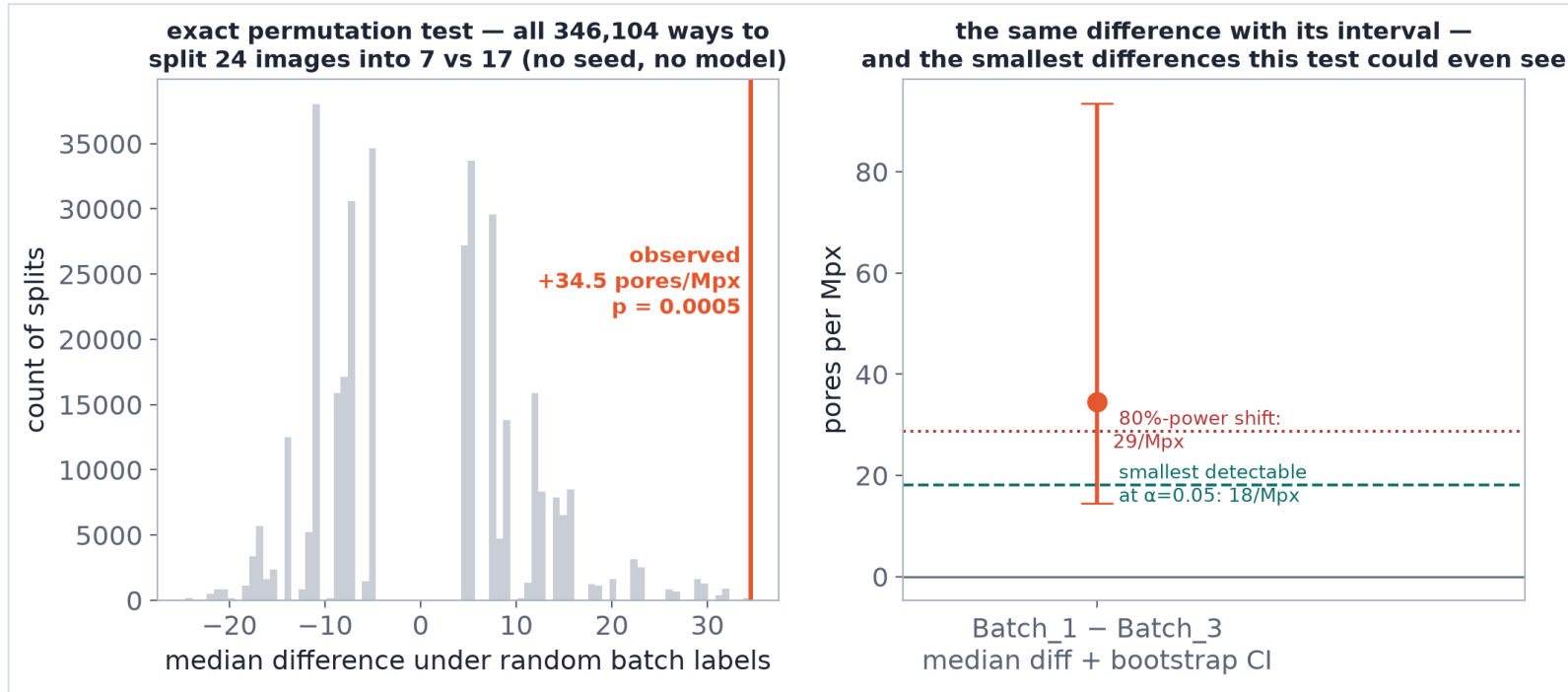
- Si fractions, sizes, clustering, porosity, anisotropy — no clear difference at these sample sizes

Read the MDD column

- differences SMALLER than the MDD would be invisible — not absent

COMPARISON

How the p-value is computed — every label shuffle



Enumerate all splits

- 346,104 ways to choose 7 of 24 images as 'Batch_1'

Rank the observed gap

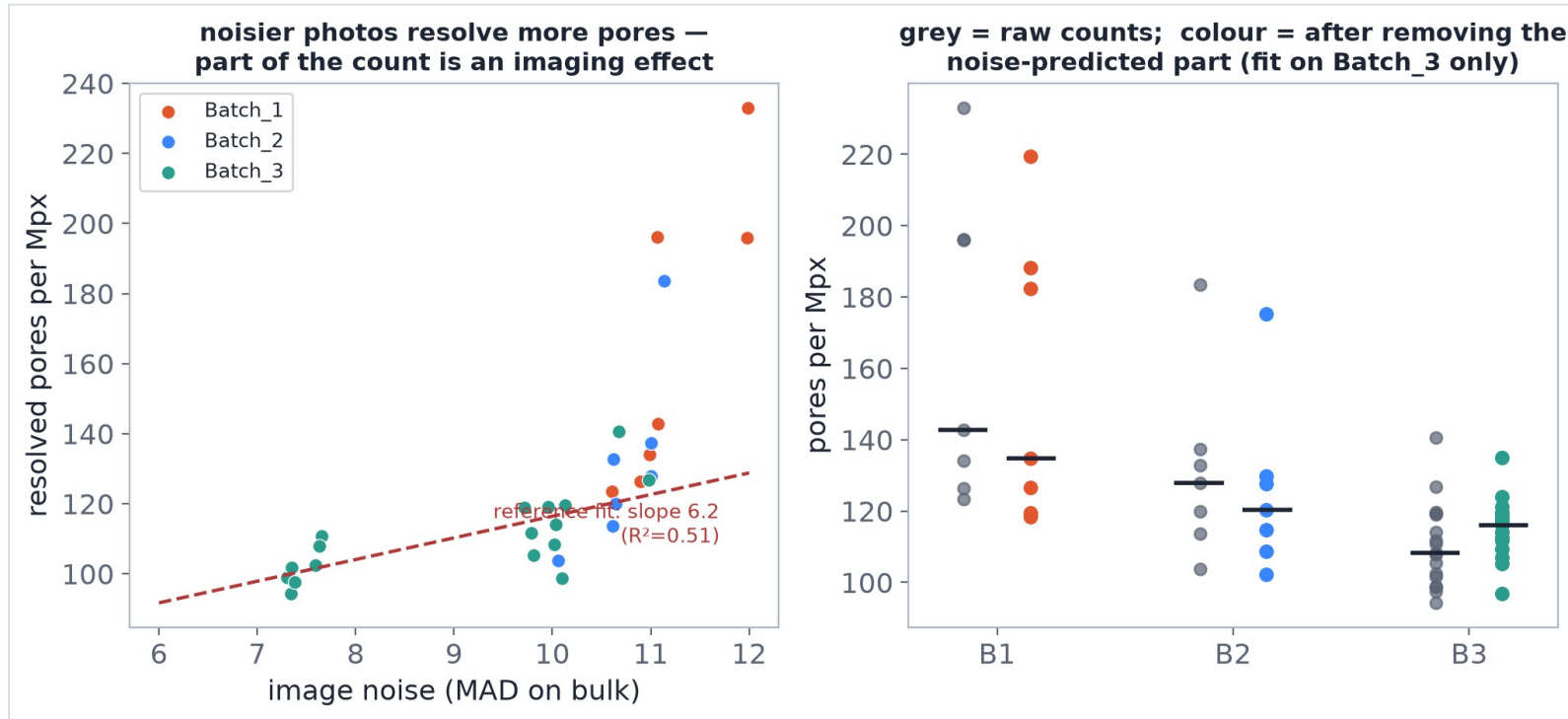
- +34.5/Mpx sits above every shuffled split $\rightarrow p = 165/346104$

Why exact

- no distribution assumed; small n handled honestly

COMPARISON

Noise inflates pore counts — adjust, then re-test



The confound

- noisier photo → more tiny threshold-crossing specks → higher pore count (correlation ≈ 0.91 with the noise_MAD guard metric)

The fix

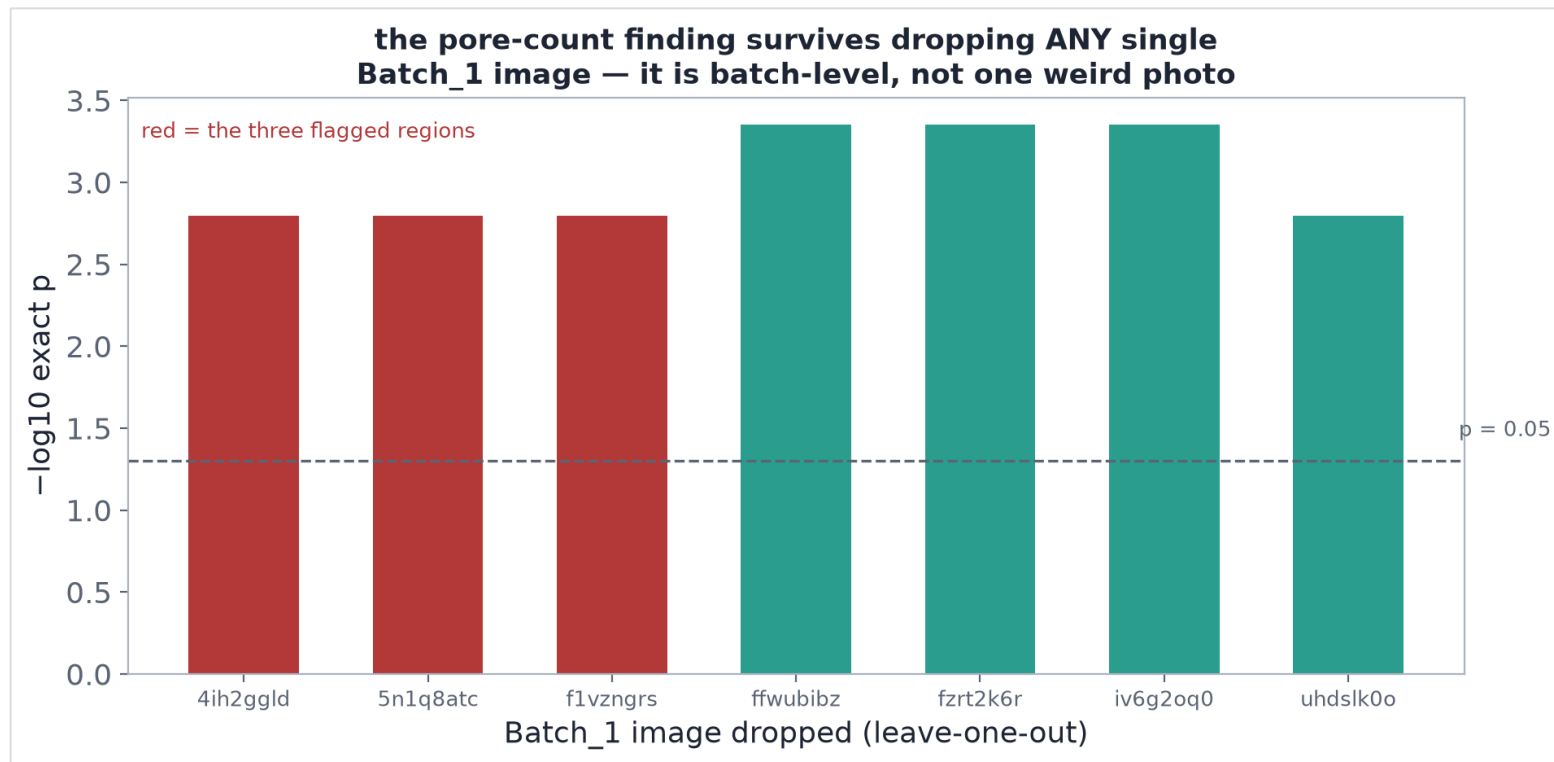
- regress pore count on image noise using REFERENCE images only (slope 6.18, fit explains $\sim 51\%$ of variation), subtract the noise-predicted part

The outcome

- Batch_1 survives (+18.8, $p=0.002$)
- Batch_2's raw shift evaporates (+4.2, $p=0.35$) — it was mostly photography

COMPARISON

Could one image be carrying the result?



Leave-one-out sweep

- drop each Batch_1 image in turn, re-run the exact test

Result

- p stays ≤ 0.0016 no matter which image is dropped
- all 7 Batch_1 images sit above the Batch_3 median — it is a batch-level shift, not one weird photo

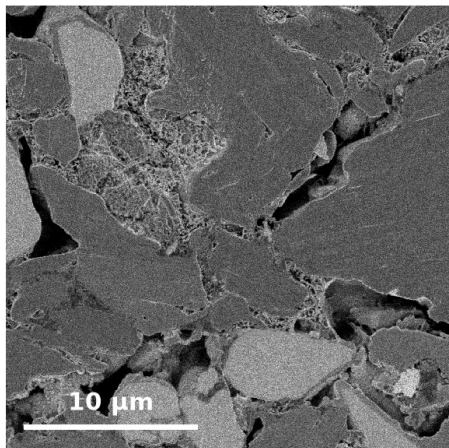
Separate question

- the 3-of-7 bright-material anomaly is a DIFFERENT, heterogeneous signal — next section

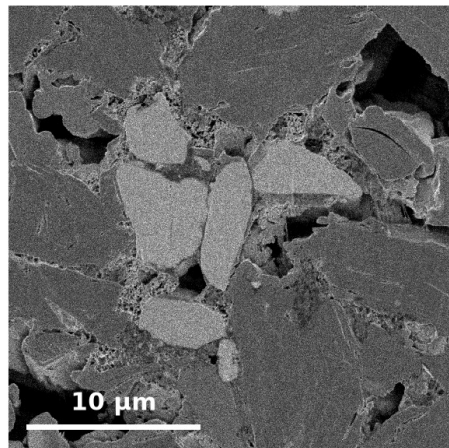
Three of seven Batch_1 regions carry extra bright material

the three flagged Batch_1 regions — extra bright material; **img_f1vzngrs** has **NORMAL** contrast

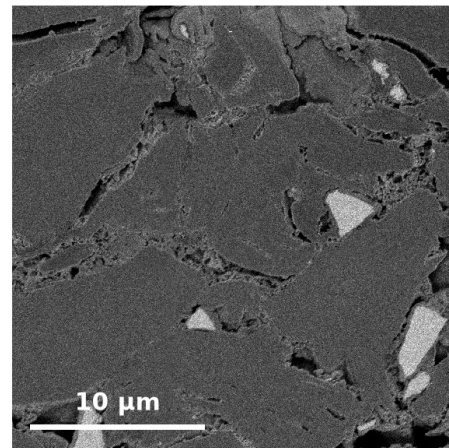
img_4ih2ggld
Si:bulk contrast=1.77 bright=12.5%



img_5n1q8atc
Si:bulk contrast=1.82 bright=11.9%



img_f1vzngrs
Si:bulk contrast=2.36 bright=10.7%



— What was flagged

- `img_4ih2ggld`, `img_5n1q8atc`, `img_f1vzngrs`
- bright material 10.7–12.5% (all-bright convention) vs baseline median 7.7%

— The complication

- 2 of 3 photos are DIMMER than baseline — their silicon glows fainter vs bulk (contrast 1.8 vs ~2.3). Real material or a contrast artefact?
- `img_f1vzngrs` shows the excess at NORMAL contrast — the interesting one

Six tests adjudicate what the bright material probably is

six tests on the disputed bright material (reconcile_output/adjudication.csv)

T1

contrast-correct the interiors

94% of ambiguous objects pass the particle cut vs 45% on baseline

→ fine Si

T2

object shape & position

solidity 0.82, ~1.05 μm discs, only 8% at image border

→ fine Si

T3

neighbourhood test

adjacent objects are compact discs (0.84), not crescent rims; they self-cluster

→ fine Si clusters

T4

re-fit thresholds per image

floating ruler counts MORE bright (20.7% vs 11.9%) — not threshold noise

→ fine Si

T5

texture on InLens & ETD

ambiguous regions have $\geq$ real-particle texture (1.20–1.23 $\times$)

→ fine Si

T6

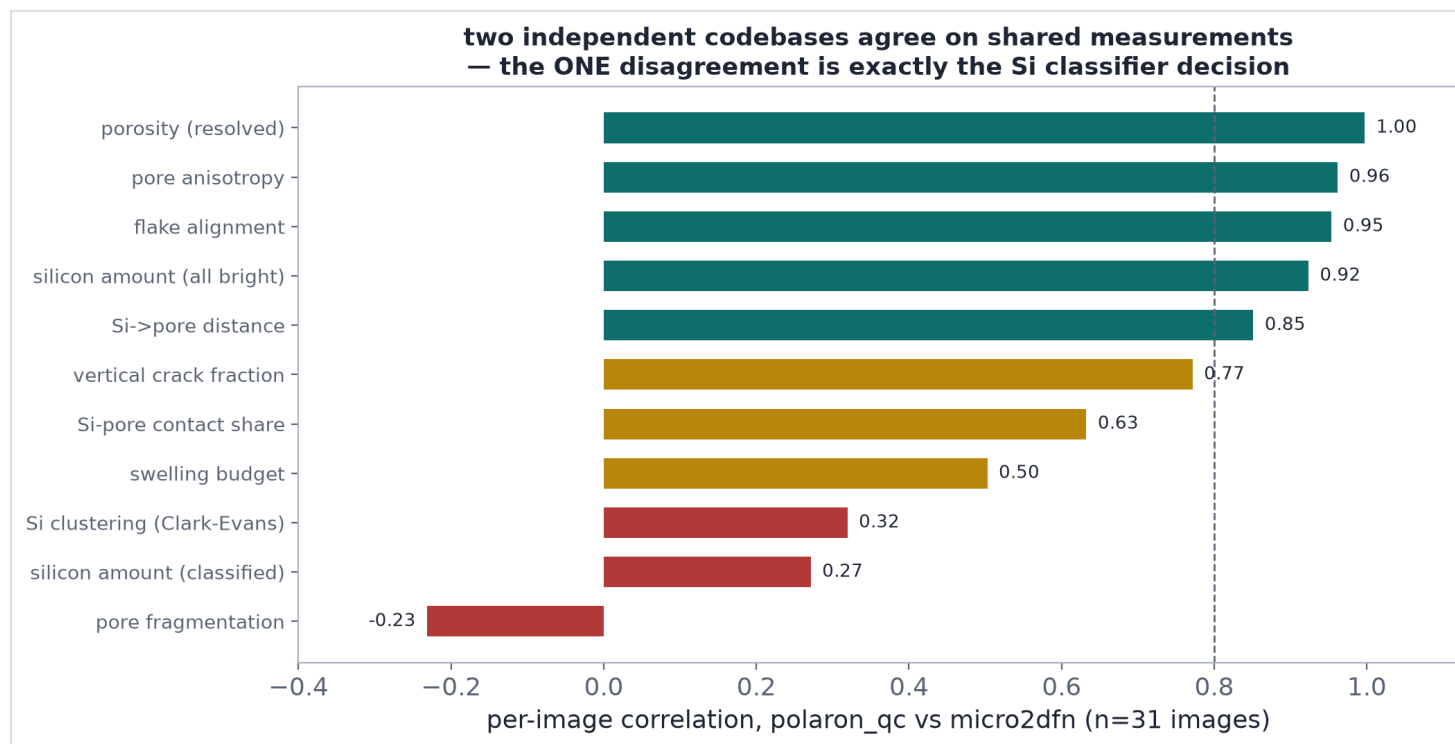
acquisition signature

whole-image median shift $z=0.0$; only the bright tail moved ($z=1.8$)

→ material change

verdict: **LIKELY FINE SILICON** — but identity still needs EDS (brightness $\neq$ chemistry); the dim contrast on 2/3 regions is a secondary anomaly

Same pixels, same flag — different word for it



Where the pipelines agree

- correlation 0.8–1.0 (1.0 = identical ranking) on raw bright fraction, porosity, anisotropy, flake alignment
- both flag the SAME three regions

Where they disagree

- classified Si only: correlation 0.27 — the classifier decision

What the verdicts mean

- REJECT = beyond the baseline's own envelope on a matched defect signature → quarantine + root-cause; NOT 'cells will fail'
- under the stricter recheck (held-out calibration) Batch_1 reads INVESTIGATE — same anomaly, stricter ruler

Where the Batch_1 evidence actually lands

— Established

- 3 of 7 Batch_1 regions carry extra bright material (both pipelines flag the same regions)
- the objects behave like fine/dimmer silicon on six pixel-level tests — LIKELY fine silicon is the best-supported reading
- batch-level resolved pore count is higher in all 7 Batch_1 images (supported comparison finding — survives noise adjustment and dropping any single image)

— Not established

- chemical identity — needs EDS/equivalent; 'silicon-family' is morphology+texture, not spectroscopy
- whether 2/3 dim-contrast photos are acquisition drift or a real dimmer-Si grade — either way it is batch non-uniformity
- any consequence for cells — no electrochemical data exist

What a DFN model is (in one picture)

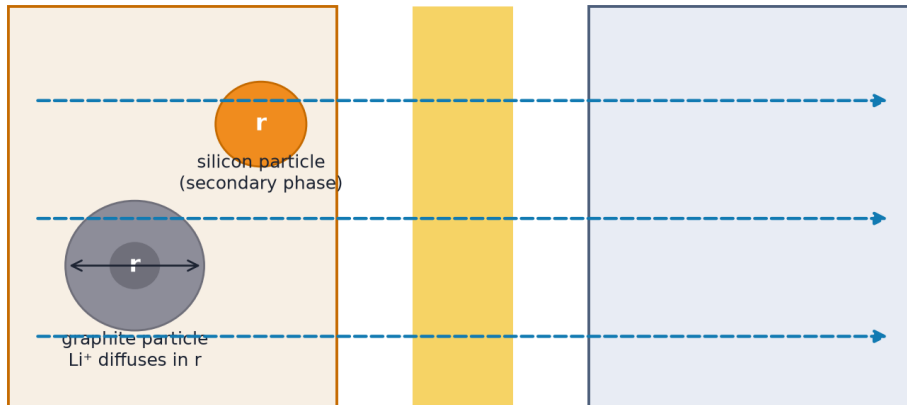
Doyle-Fuller-Newman: the electrode as linked continua — not individual particles

electrons travel in the solid, opposite way

ANODE (this is what we image)

separator

CATHODE (NMC)



x — Li^+ travels through electrolyte in the pores

the model solves, at every x and t :

- Li^+ concentration in electrolyte
- Li inside each particle (r)
- potentials of solid & electrolyte
- SEI growth
- particle swelling & stress

SCHEMATIC — drawn for teaching, not measured

— Doyle-Fuller-Newman

- two porous electrodes + separator; lithium tracked through electrolyte and inside solid particles

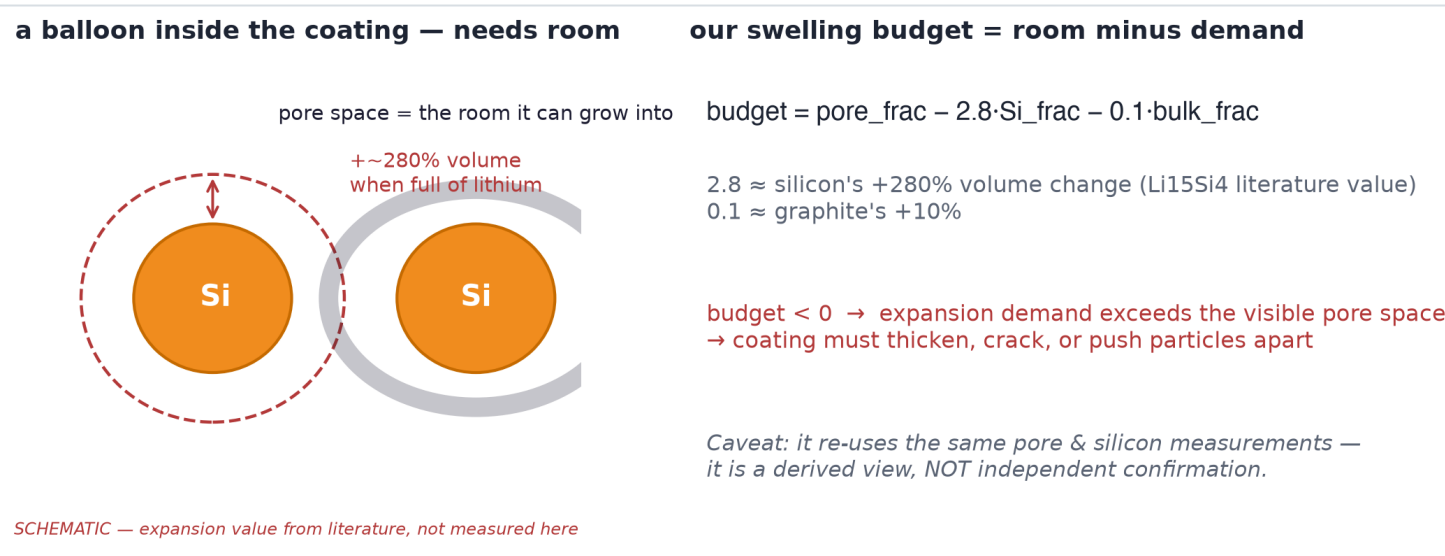
— Our two-material negative

- primary phase = graphite; secondary phase = silicon with swelling + stress coupling

— Why bother

- it turns 'more/finer/coarser structure' into signed changes in model indicators — under declared assumptions

Why silicon is special: it swells ~3x on lithiation



— The numbers

- Si: ~+280% volume at full lithiation (Li15Si4)
- graphite: ~+10%

— The consequences in the model

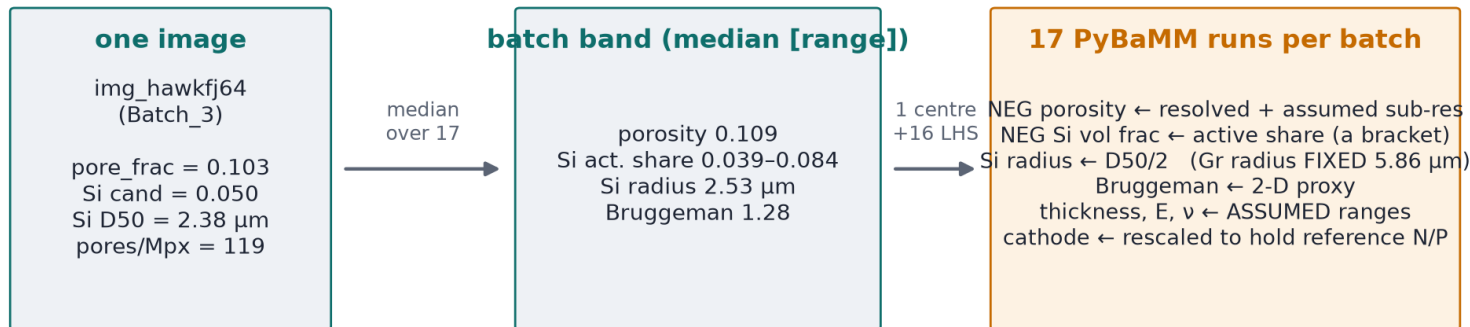
- particle stress → mechanical fade coupling (stress-LAM = stress-driven loss of active material)
- electrode-thickness growth
- pore space is the expansion buffer → the 'swelling budget' proxy

— The image link

- Si fraction + resolved porosity set how much buffer the structure shows — within the resolution floor

Which measurements feed which model inputs

tracing one field of view into the model



*an image never becomes a cell: per-image markers → batch medians → ONE point inside an assumption band
→ 17 model runs that sweep the unmeasured inputs*

pore count is NOT among the inputs — the pore finding lives outside this model entirely

— Measured → model

- resolved porosity → porosity (plus assumed sub-res add-on)
- Si D50/2 → Si particle radius (swept over the batch band)
- classified Si x accessible share → Si active fraction, LOW end of bracket
- ALL uncertain bright counted → HIGH end of bracket

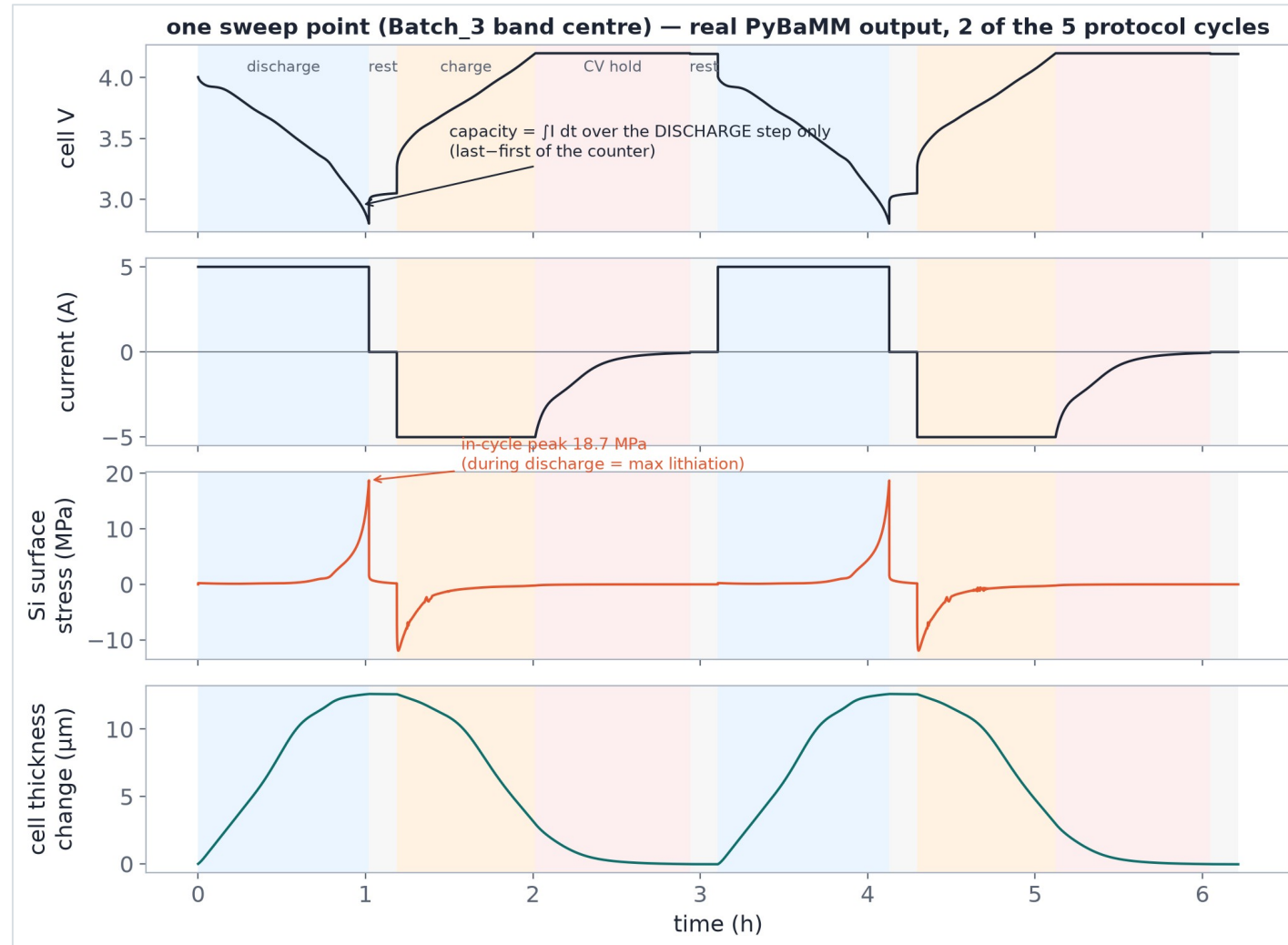
— Assumed → swept

- thickness 65–85 μm; sub-res porosity +0.15–0.30; Si modulus 35–90 GPa; nu 0.22–0.28

— Not a model input

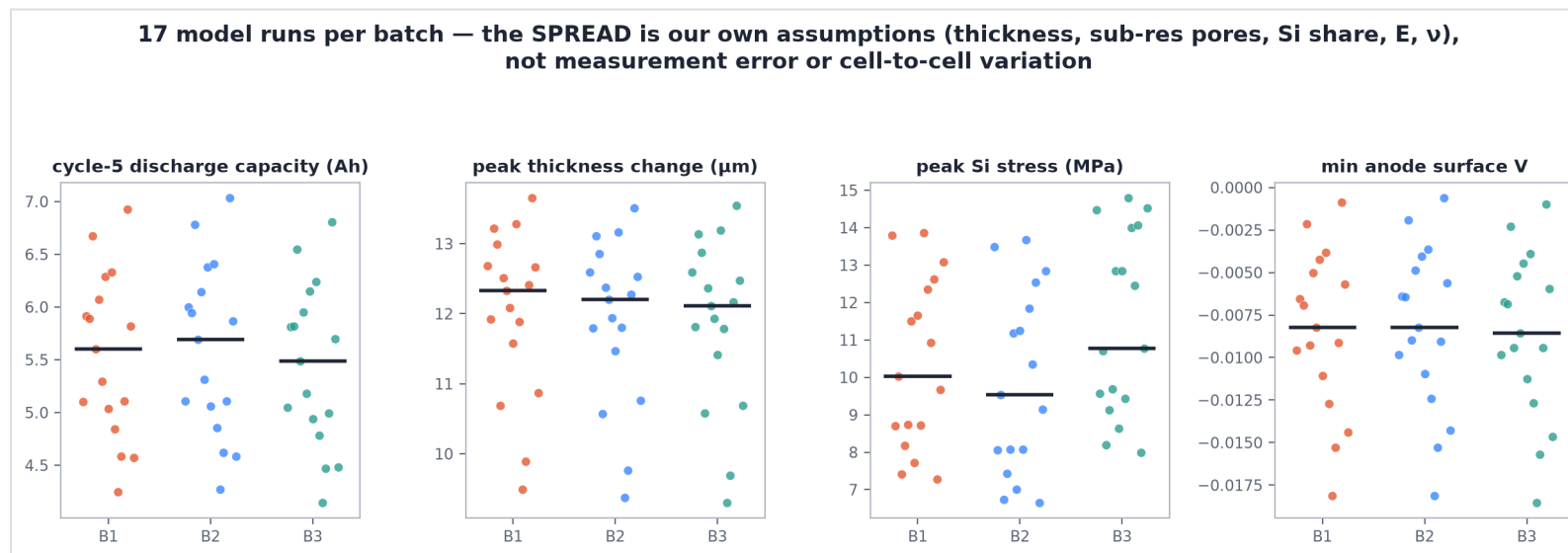
- pore COUNT — the one supported finding is invisible to the DFN entirely

One sweep run, fully real: the protocol and the indicators



Actual PyBaMM solution at the Batch_3 band centre — 5 x (1C discharge → rest → 1C charge → CV hold → rest). 1C = a full charge/discharge in ~1 hour; CV hold = constant-voltage top-up. Indicators read as in-cycle peaks/deltas.

The assumption sweep: 17 runs per batch



Why sweep

- unmeasured inputs are bands, not points — the honest output is a band of indicators

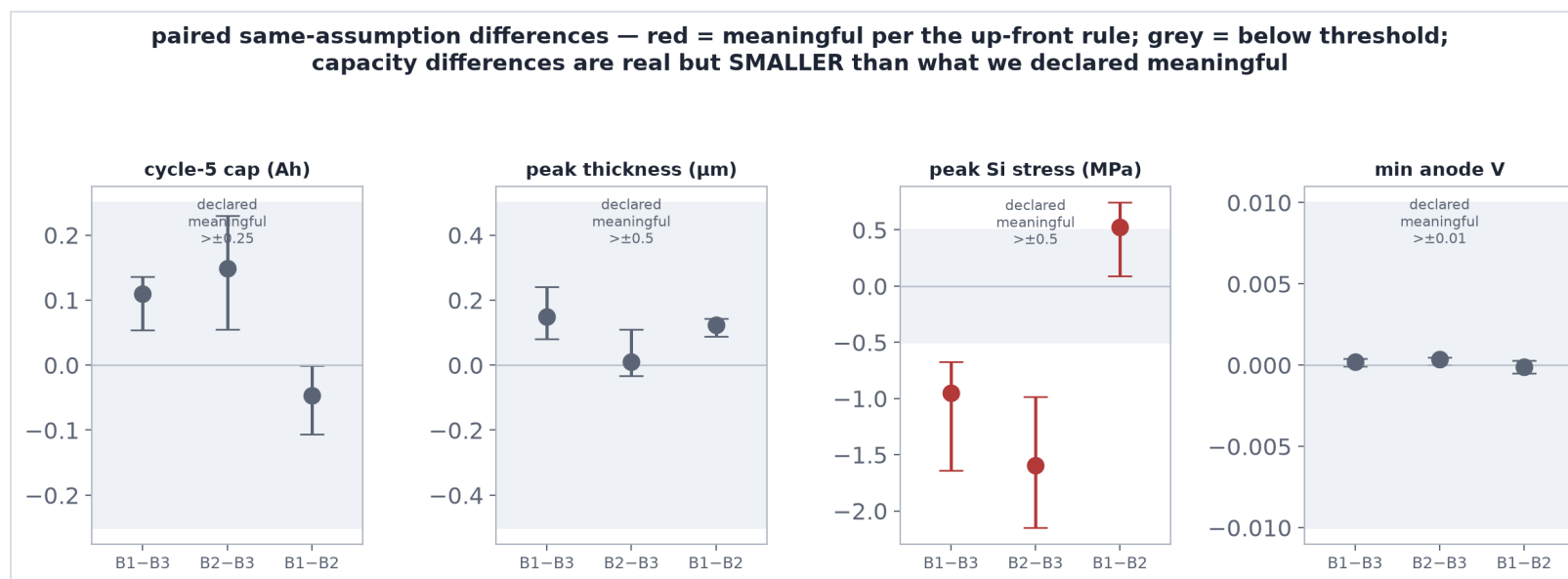
17 points

- band centre + corners + edges over porosity / Si share / radius / mechanics / thickness

The picture

- batch medians sit inside overlapping bands — the assumptions dominate the differences

Paired differences: what moves, what doesn't



Declared thresholds first

- capacity 0.25 Ah (amp-hours) · swelling 0.5 μm · stress 0.5 MPa (megapascals) · min anode voltage 0.01 V — fixed before looking

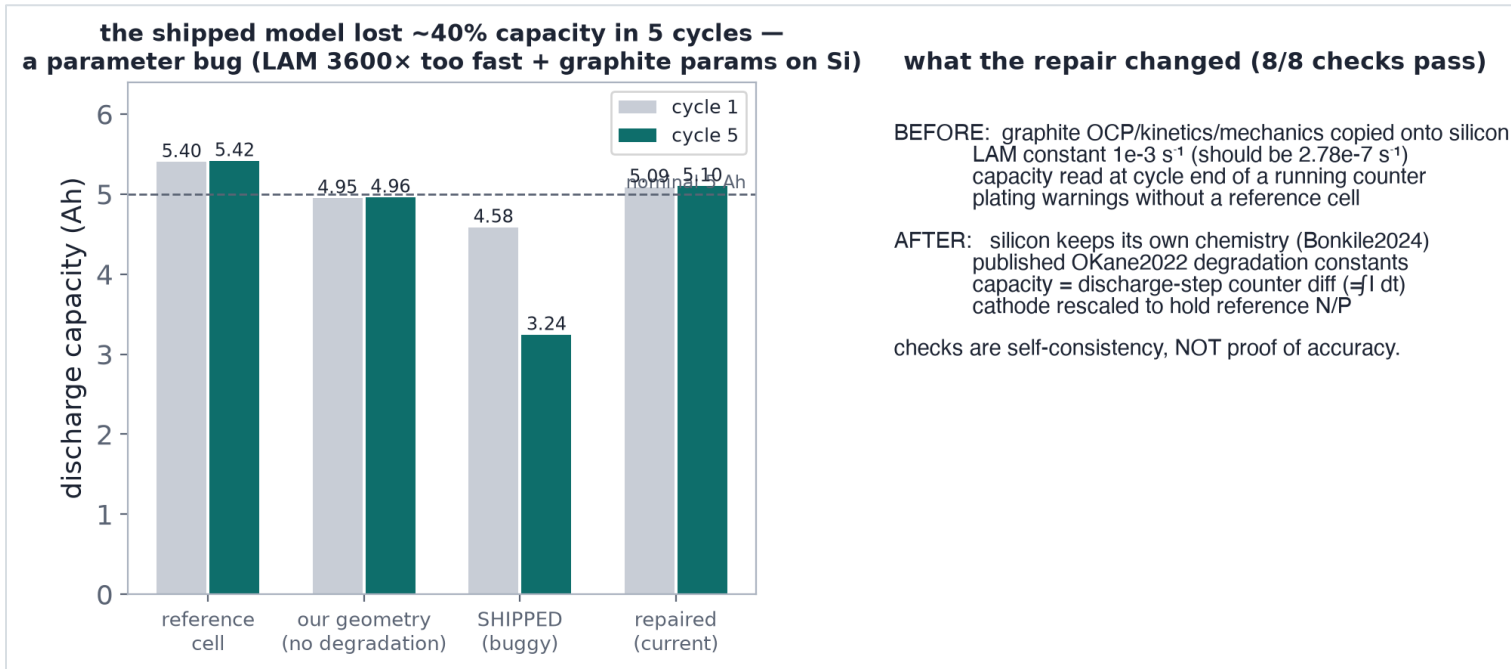
The reads

- capacity: B1-B3 +0.099, B2-B3 +0.150 Ah — below threshold
- Si stress: B1-B3 -0.81, B2-B3 -1.59 MPa — crosses the declared bar
- swelling & plating-margin: below threshold

The catch

- the stress shift rides on assumed Si share + modulus — model-conditional, not measured

Validation means self-consistency — and it caught real bugs



— What was wrong

- shipped constants had the fade rate (LAM = loss of active material) 3600x too fast and the protective-film rate (SEI = solid electrolyte interphase) 1000x too slow vs the cited sources

— What it did

- buggy model lost ~29% capacity in 5 cycles — physically absurd, and it would have shipped

— What validation is

- 8/8 self-checks now pass — reference sanity, ablation, convergence, protocol audit

— What it is NOT

- no experimental data exist → nothing here validates predictive accuracy

Where the evidence stops — the honest ledger

— Established (measured)

- Batch_1 pore-count shift vs Batch_3 (uniform, robust)
- 3-of-7 bright-material anomaly; likely fine silicon
- both pipelines detect; they differ only on naming

— Best-supported but open

- stress-indicator differences (model-conditional)
- dim-contrast on 2 flagged photos (drift or grade?)

— Not established (can't be, with this data)

- chemical identity · any cell-level consequence
- general batch ranking · equivalence on the other metrics

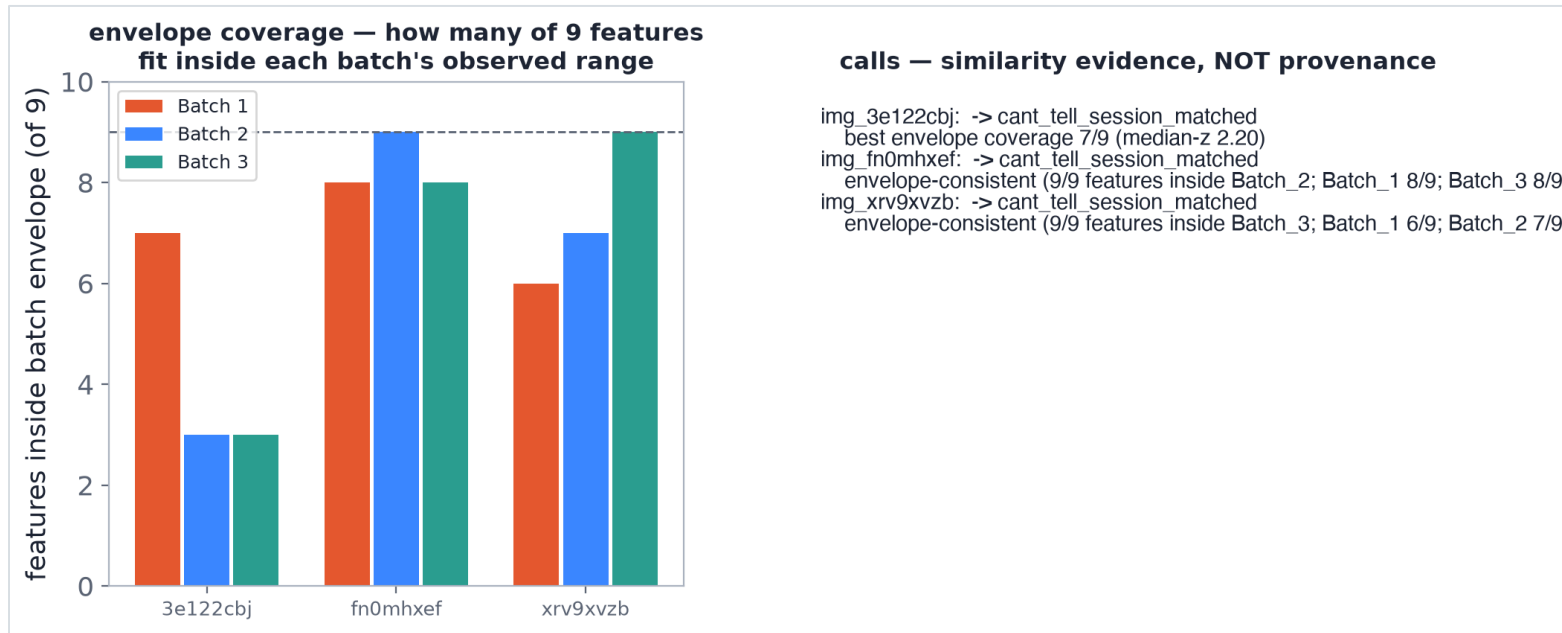
— The honest ledger

- assumed inputs: thickness, sub-res porosity, Si share bracket, mechanics, degradation constants
- assumption spread > every batch difference on capacity
- 34 images is small — MDDs bound what 'no difference' means

— What would close the gaps (roadmap.md)

- EDS (elemental spectroscopy) on flagged regions → chemistry
- controlled repeat imaging → acquisition effects
- blind expert annotation → labels for classifier eval
- measured thickness/porosity → shrink the assumed bands
- electrochemical testing → ground the indicators

Three unlabelled photos: similarity, not provenance



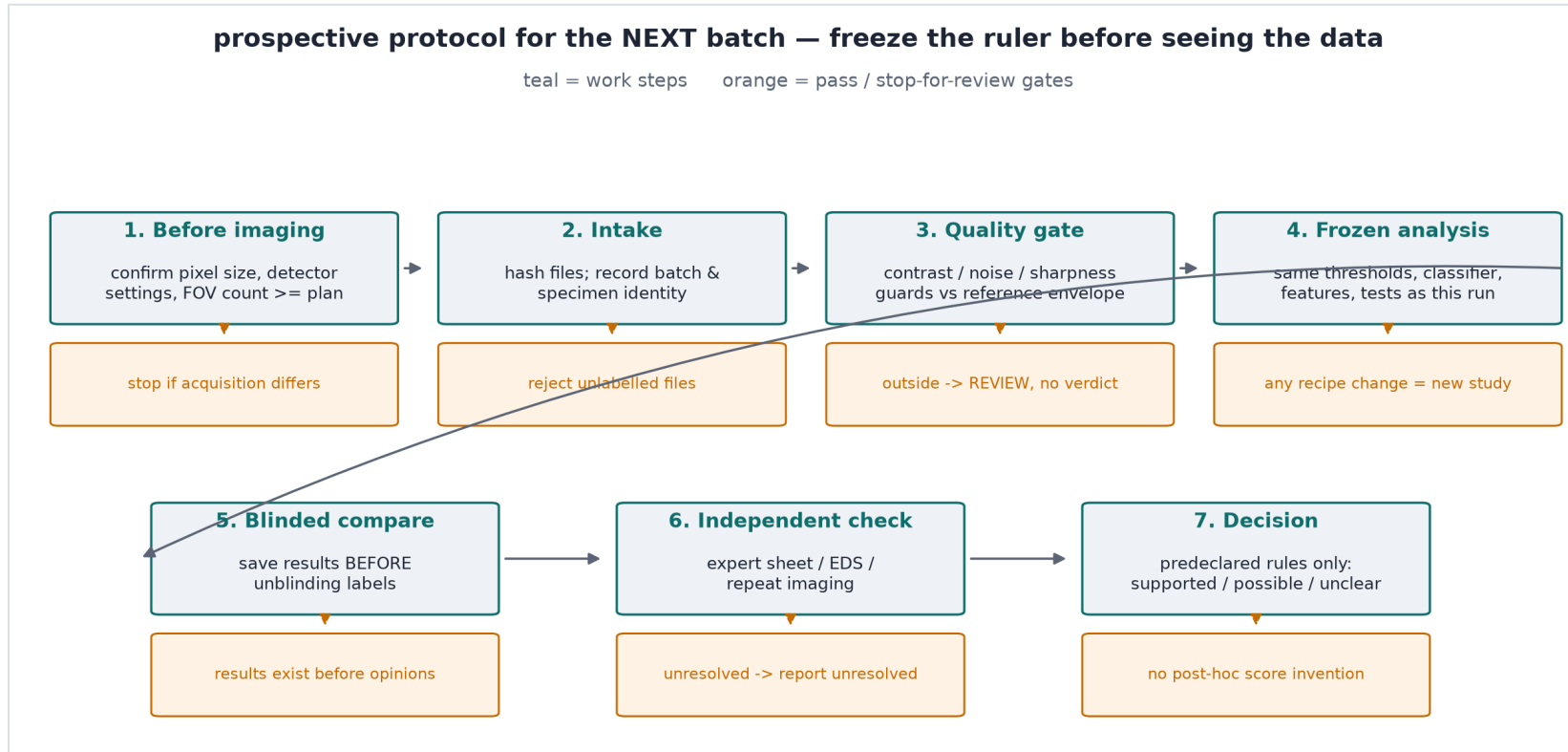
Envelope coverage on the 9 features

- img_3e122cbj → inside Batch_1 envelope on 7/9 — including the uncertain-bright signature → assigned Batch_1 (cautious)
- img_fn0mhxef → 9/9 inside Batch_2
- img_xrv9xvzb → 9/9 inside Batch_3

Low confidence by design

- overlapping envelopes = similarity classes, not identity; the labels are hypothesis-generating only

If this were run prospectively: the frozen protocol



— Freeze BEFORE new data

- recipe, reference, classifier, model assumptions, comparisons, decision rules — all versioned

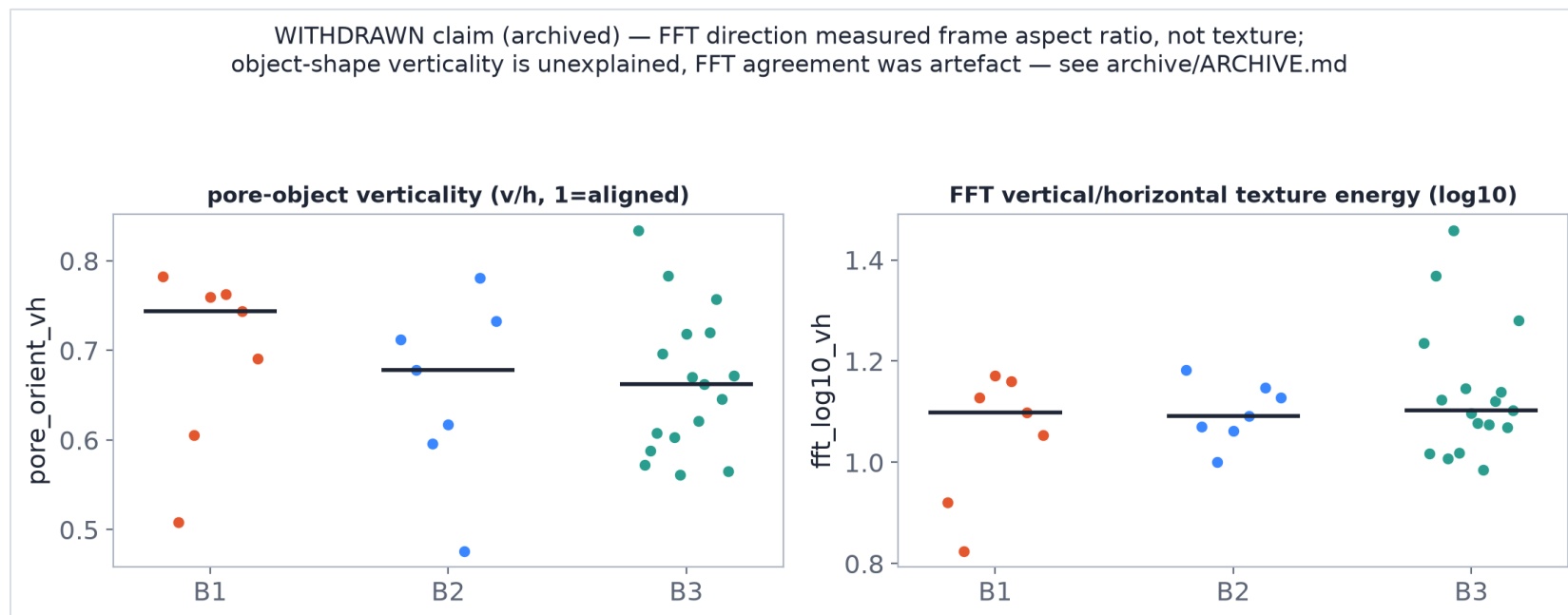
— Gates that stop the line

- acquisition outside supported conditions → review, don't force a verdict
- image-quality guards fail → investigate imaging first

— Unblind last

- save results before revealing labels; evaluate at the image unit

Bonus open question: through-plane pore texture



What we claimed

- pores and texture prefer the vertical/through-plane direction (FFT directionality + object shape)

What the audit found

- the FFT score measures the frame's aspect ratio — pure noise scores the same at the real 1034x3500 frame shape
- direct autocorrelation shows features ~1.4x HORIZONTAL — the claim was withdrawn

Why show a retracted result

- this is what honest reporting looks like: the finding was in orientation_features.csv for weeks before anyone ran the noise control

Five things to keep

- **1. Labels are model outputs.** Every 'material fraction' is a threshold+classifier choice — freeze and version the recipe.
- **2. The image is the unit.** 34 images is small; 'no clear difference' carries a minimum-detectable-difference, never equivalence.
- **3. One supported finding.** Batch_1 > Batch_3 on noise-adjusted resolved pore count (uniform, robust to dropping any image) — plus a separate 3-of-7 likely-fine-silicon anomaly still awaiting chemical confirmation.
- **4. Model outputs are conditional.** DFN (battery-physics) indicators move with assumed inputs more than with batches — report paired differences against declared thresholds, never absolute claims.
- **5. Verdicts are similarity statements.** ACCEPT/INVESTIGATE/REJECT describe distance from a SELECTED reference — not chemistry, performance, or safety.

Traceability and further reading

— Trace your numbers

- `claim_evidence.csv` — 50 claims → source file + class
- `metric_dictionary.csv` — units, equations, limits
- `dataset_manifest.csv` — every image + md5

— Self-test

- `knowledge_check.md` — 8 questions with answers

— Rebuild everything

- `BUILD.md` — one command regenerates figures + deck

— Expert review

- `expert_review_sheet.csv` — what a human expert should verify and sign off

— Primary sources

- Doyle, Fuller, Newman — JECS 140:1526 (1993): the DFN model
- Chen et al. — PhysChemChemPhys 22 (2020): parameter set
- OKane et al. — PhysChemChemPhys 24 (2022): degradation constants
- Bonkile & Ramadesigan — J. Power Sources 581 (2024): Si mechanics
- Otsu — IEEE SMC-9 (1979); Saltykov stereology; Delesse (1848); Clark & Evans (1954); Benjamini & Hochberg JRSS-B 57 (1995)

All 9 features x 3 batch pairs (permutation + BH)

B1-B2

feature	media n diff	exact p	BH	MDD(8 0%)
si_candidate_frac	-0.00187	0.6503	—	0.0126
uncertain_bright_frac	-0.00333	0.6503	—	0.0116
si_d50_um	-0.138	0.6503	—	0.59
si_d90_um	-0.558	0.6037	—	1.83
si_clustering_R	-0.016	0.1841	—	0.0464
pore_frac	-0.00453	0.7669	—	0.0288
pores_per_mpx	+15	0.2191	—	65.3
pores_per_mpx_noise_adj	+14.6	0.2191	—	64.5
pore_anisotropy	+0.0162	1	—	0.083

B1-B3

feature	media n diff	exact p	BH	MDD(8 0%)
si_candidate_frac	-0.000803	1	—	0.0236
uncertain_bright_frac	-0.00283	0.764	—	0.0153
si_d50_um	+7.44e-05	1	—	0.548
si_d90_um	+0.0544	0.6854	—	1.04
si_clustering_R	+0.0118	0.6138	—	0.0752
pore_frac	-0.00516	0.5967	—	0.0312
pores_per_mpx	+34.5	0.0004767	YES	28.7
pores_per_mpx_noise_adj	+18.8	0.001864	YES	10.3
pore_anisotropy	-0.0208	0.5829	—	0.0839

B2-B3

feature	media n diff	exact p	BH	MDD(8 0%)
si_candidate_frac	+0.00107	0.5786	—	0.0148
uncertain_bright_frac	+0.000509	1	—	0.00973
si_d50_um	+0.138	0.5071	—	0.469
si_d90_um	+0.612	0.3229	—	1.46
si_clustering_R	+0.0278	0.5256	—	0.0975
pore_frac	-0.000629	1	—	0.0267
pores_per_mpx	+19.5	0.01007	YES	21.6
pores_per_mpx_noise_adj	+4.16	0.3528	—	12
pore_anisotropy	-0.0369	0.1675	—	0.0778

validated_comparison/pairwise_comparisons.csv — the authoritative table; 'no clear difference' rows carry MDDs.

Corrected calibration: what changed and why it matters

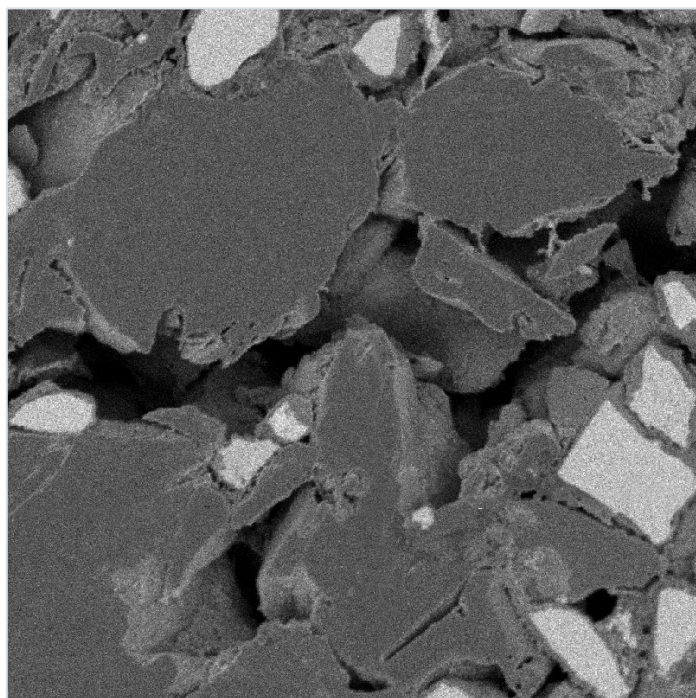
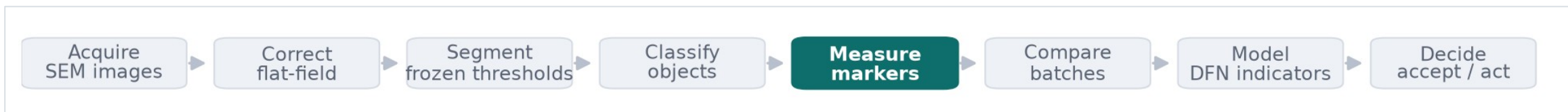
- **Original QC verdicts:** Batch_1 REJECT · Batch_2 INVESTIGATE
 - the flag weights + T^2 reference (a multivariate 'how-unusual' statistic) were fitted on data that saw the batches under test
- **Corrected recheck** (reference-only weights + held-out T^2): BOTH batches → INVESTIGATE
 - Batch_1: 1/7 regions exceeds flag threshold; 2 more above the baseline's p90 (90th percentile, same signature); held-out T^2 unusual ($p \approx 0.00005$)
- **Baseline honesty check:** LOO exceedance = $1/17 \approx 5.9\%$ [CI 1.0–26.9%] — the earlier 'zero false alarms' claim is withdrawn
- **Standing today:** the anomaly EXISTS; the severity label is calibration-dependent → INVESTIGATE is the defensible call

The declared assumptions (full sweep table)

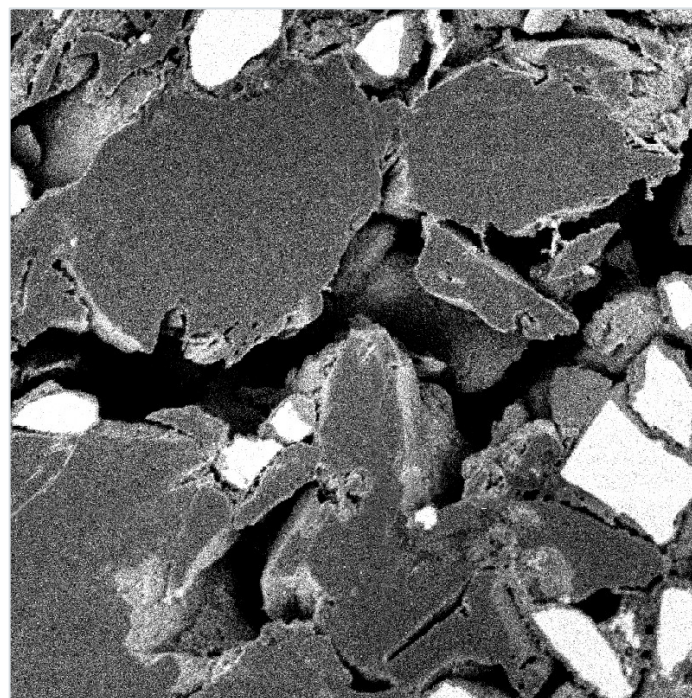
input	value	why assumed
electrode thickness	75 um [65-85 swept]	foil & coating top not in frame
sub-resolution porosity	+0.20 [+0.15 to +0.30]	pores below ~0.35 um are invisible
Si active share	0.04 [0.04-0.08]	lo: classified Si x share touching pore; hi: ALL uncertain bright counted (label bracket); 'all-Si-active' var
Si Young's modulus	50 GPa [35-90 swept]	Bonkile2024; lithiated-Si literature range
Si Poisson's ratio	0.25 [0.22-0.28 swept]	Bonkile2024; cited range
Si particle radius	1.26 um	d50/2 of detected Si (area-weighted)
graphite particle radius	5.86 um FIXED, identical for every batch	fixed assumed value (Chen2020) — NOT measured from images
pore count / pore density	not a DFN input	resolved-pore-count differences (e.g. Batch_1 vs Batch_3) are image findings only — the homogenised model has
primary-phase chemistry	graphite (NMC622 cathode)	Chen2020_composite baseline cell
Bruggeman exponent	1.28	2-D finite-difference tortuosity proxy
degradation constants	OKane2022 (LAM prop. 2.7778e-7 s-1, SEI kinetic 1e-12 m/s)	published; see dfn_output/dfn_validation.md constants table

dfn_output/assumption_table.csv — every unmeasured input is declared with its source and sweep band.

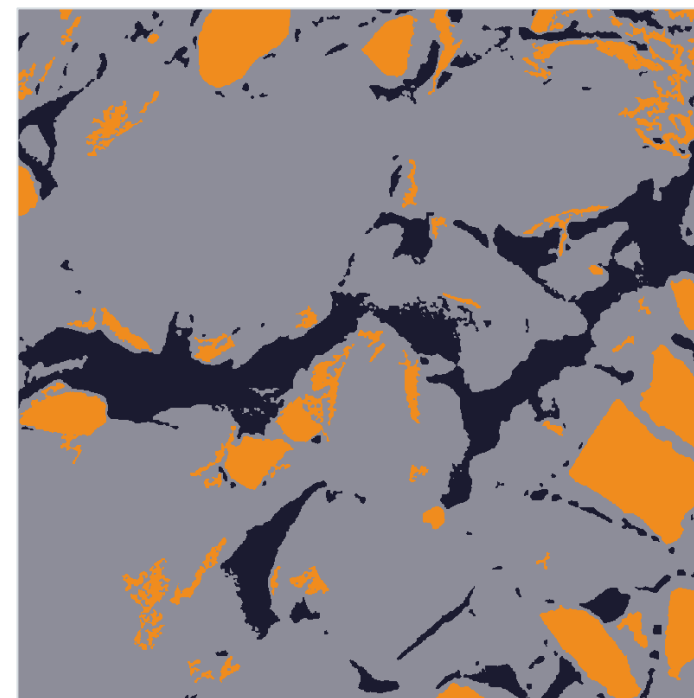
The transformation chain on the reference crop



raw



flat-fielded



labelled

Reproduce it: environment, seeds, commands

— Environment (version_manifest)

- Python 3.13.5 · NumPy 2.5.3 · pandas 3.0.6 · SciPy 1.18.1 · scikit-image 0.26.0 · matplotlib 3.11.2 · PyBaMM 26.9.0.0

— Self-updating pack

- scripts/sync.py fingerprints all inputs, rebuilds only changed stages, and re-audits every claim
- build stamp — sync 2026-10-04T02:44:02Z · 144 fingerprinted inputs · stages: figures, deck, pdf, render

— Seeds and recipes

- comparison recipe vcompare-1.0, seed 0, fitted on 17 Batch_3 images (recipe.json, md5'd inputs)

— Rebuild this teaching pack

- scripts/sync.py — one command does manifest → figures → deck → PDF → claim audit
- audit_claims.py — re-verifies every file-checkable claim; STALE = source no longer matches the deck

— Traceability

- claim ids (C01–C50) in speaker notes ↔ claim_evidence.csv